08/10/26

Signal Processing

- Recap: z-transform properties; ROC needs attention
- "Modulation" or Multiplication
- Convolution
- $x_1[n] = z_0^n \cdot x[n]$

Recall $e^{j\omega_0 n} \cdot x[n]$.

Q. If $x[n] \longleftrightarrow X(z)$ with ROC R_x, what is the z-transform of $x_1[n]$?

A: $X_1(z) = X(z/z_0)$ with ROC equal to $|z_0| \cdot R_x$.

$$X_1(z) = \sum_{n=-\infty}^{\infty} x_1[n] \cdot z^{-n}$$

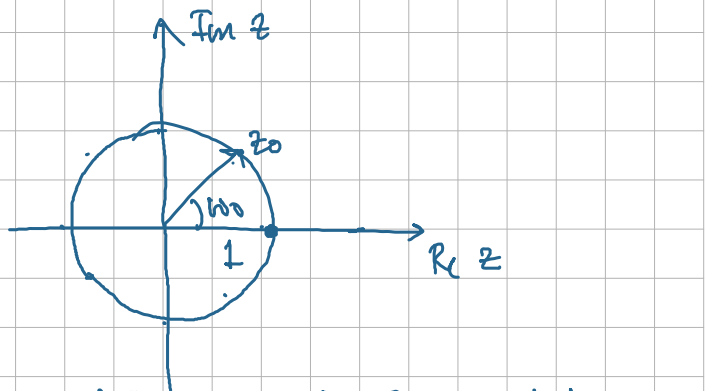
$$= \sum x[n] \cdot z_0^n \cdot z^{-n}$$

$$= \sum_{n=-\infty}^{\infty} x[n] \underbrace{(z/z_0)^{-n}}_{z'}$$ ← converge for all $z/z_0 \in R_x$

in other words $z \in |z_0| R_x$

$= X(z')$ with the ROC being $|z_0| \cdot R_x$.

If we want to interpret the modulation result $e^{j\omega_0 n} x[n]$, we can think of the case $z_0 = 1 \cdot e^{j\omega_0}$.



- Convolution: If $x_1[n] \longleftrightarrow X_1(z)$, $x_2[n] \longleftrightarrow X_2(z)$. R_{x_2}, what is the z-transform of $x[n] = x_1[n] \overset{R_{x_1}}{*} x_2[n]$?

$$X(z) = X_1(z) \cdot X_2(z). \text{ with } R_x = \underbrace{R_{x_1} \cap R_{x_2}}$$

Recall: $Y(z) = H(z) \cdot X(z)$